

in the Universe, hydrogen had been detected and burned by earlier scientists, but Cavendish was the first to recognize it as a distinct substance. He also isolated carbon dioxide, which was then known as “fixed air.” He compared hydrogen and carbon dioxide’s chemical and physical properties with those of ordinary air. To determine hydrogen’s density, he filled a pig’s bladder with it and, weighing that, calculated the gas to be 11 times lighter than air (slightly less than the modern accepted figure). He found carbon dioxide to be 1.57 times heavier than air and dissolvable in water.

Measuring water and air

Cavendish then turned to exploring the composition of ordinary, or atmospheric, air. He analyzed air on 60 occasions and from different places, even using a balloon to collect samples from the atmosphere. He expressed his results in terms of Johann Becher’s phlogiston theory, which stated that all combustible matter contained an odorless and colorless substance called “phlogiston.” When matter was burned, the phlogiston was released, after which the remaining matter, now ash or residue, was considered to be “dephlogisticated.”

Correctly reporting that air is four parts “phlogisticated air” (nitrogen) to one part “dephlogisticated air” (oxygen), Cavendish also found a scant amount of a gas that he could not identify. A century later, this was identified as argon.

Having observed that burning his inflammable air (hydrogen) also produced water, in 1784, Cavendish published a paper that revealed water’s chemical formula—two parts inflammable air (hydrogen) to one part dephlogisticated air (oxygen). He had proved that water is

“He was not a Poet,
a Priest, or a Prophet,
**but only a cold,
clear Intelligence.”**

George Wilson, 1851

not a chemical element—a single, distinct substance—as had been thought for millennia, but a compound of two gases.

Cavendish experiment

Between 1797 and 1798, Cavendish devised an ingenious experiment to determine the density of the Earth—an elusive value sought by both scientists and explorers. His apparatus for “weighing the world” was a wooden rod suspended from a wire, with a small lead sphere on either end and two much larger lead spheres fixed in place close to the smaller spheres. Since all objects with mass exert a gravitational force on each other, the large spheres attracted the smaller ones, causing a slight torque (twist) on the suspended



Cavendish's experiment, which ultimately enabled him to calculate the density of the Earth, also proved that Newton's law of gravitation works with objects much smaller than the planets.